Algebraic Number Theory

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Disclaimer

This course is presented by Pr. CAMPAGNA Francesco and Pr. STEVEN-HAGEN Peter.

To separate the contents of the course to actual additions or out of context information, a black band will be added by its side like the one on this comment.

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Chapter 1

Introduction

- **Problem 1:** Find all prime numbers $p \in \mathbb{N}$, such that $\exists x, y \in \mathbb{Z}, p = x^2 + y^2$. For small values we have

p	(x, y)
2	$(1, 1)$
3	✗
5	$(1, 2)$
7	✗
11	✗
13	$(2, 3)$

We have the following observation, if $p \neq 2$ then $p = x^2 + y^2$ then $p \equiv 1 \bmod 4$, which is true since $x^2 + y^2 \equiv 0, 1, 2 \bmod 4$ and since p is an odd prime then necessarily $p \equiv 1, 3 \bmod 4$ hence $p \equiv 1 \bmod 4$.

Now the question is if the other implication, if $p \equiv 1 \bmod 4$ then is it necessary that $p = x^2 + y^2$. If we consider the equation $p = x^2 + y^2$ in the Gaussian integers $\mathbb{Z}[i]$, then we have that $p = (x + iy)(x - iy)$. $\mathbb{Z}[i]$ is a Euclidean domain (if we consider $\alpha = \beta q + r$, we can rewrite it as $|\alpha/\beta - q| = |r/\beta| < 1$, the disks with radius 1 around each points of the lattice $\mathbb{Z}[i]$ cover the whole space) and thus it is a principal ideal domain, so α factors into irreducibles. Looking at $\mathbb{F}_p^\star$, we have that $-1 \in \mathbb{F}_p^\star$ has order 2, if $p \equiv 1 \bmod 4$ then there exists an element $x \in \mathbb{Z}, x^2 \equiv -1 \bmod p, p \mid x^2 + 1 = (x + i)(x - i)$ in $\mathbb{Z}[i]$ but $p \nmid x + i$ and $p \nmid x - i$ thus p is not irreducible in $\mathbb{Z}[i]$, thus $p = \pi \cdot \pi'$ with π irreducible, we have $p = N(p) = N(\pi \cdot \pi') = N(\pi)N(\pi') \Rightarrow N(\pi) \mid p^2$, if $N(\pi) = 1$ then $\pi \in i^{\mathbb{N}}$ which is not the case, hence $N(\pi) \neq 1$ and the same for $N(\pi')$, thus $N(\pi) = p$, since $\pi = a + bi$ then $p = N(\pi) = a^2 + b^2$.

- **Problem 2:** Find all integer solutions of the equation $y^3 = x^2 + 1$. By considering it on $\mathbb{Z}[i]$ we have the equivalent equation $y^3 = (x - i)(x + i)$, $(x - i)$ and $(x + i)$ are coprime given that if there is $\pi \in \mathbb{Z}[i]$ such that it divides them both, then it divides the difference $\pi \mid 2i \Rightarrow \pi \mid 2$, and thus from the equation $2 \mid y \Rightarrow 8 \mid y^3$ thus $x^2 \equiv -1 \bmod 8$ which is impossible, thus $x - i$ and $x + i$ are coprime. In an UFD, if a, b coprime and $ab = \gamma^2$ then a, b are squares up to a unit, we have that $x + i = (a + bi)^3 = a^3 + 3a^2bi - 3ab^2 - b^3i = (a^3 - ab^2) + (3a^2b - b^3)i \Rightarrow x = a^3 - ab^2, 1 = 3a^2b - b^3$, from the second equation we have that necessarily $b^2 = 1$, replacing it back we have that if $b = 1$ then $3a^2 - 1 = 1 \Rightarrow a^2 = \frac{2}{3}$ which is impossible, if $b = -1$ then $a = 0$ and thus $x = 1$, therefore, the only solutions there are in this case is $(x, y) = (0, 1)$.

- **Problem 3:** Solve the Diophantine equation $y^3 = x^2 + 19$. By doing the same process as earlier, we have that $y^3 = \left(x + \sqrt{-19}\right)\left(x - \sqrt{-19}\right)$ in $\mathbb{Z}\left[\sqrt{-19}\right]$ which is a number ring, lets imagine that $\mathbb{Z}\left[\sqrt{-19}\right]$ has the same properties as $\mathbb{Z}[i]$, by the same way we can have that $x - \sqrt{-19}$ and $x + \sqrt{-19}$ are coprime, thus we deduce that $x + \sqrt{-19} = \left(a + b\sqrt{-19}\right)^3 u$, we define the norm $N : \mathbb{Z}\left[\sqrt{-19}\right] \to \mathbb{N}, a + b\sqrt{-19} \mapsto a^2 + 19b^2$, we have that the units satisfy that $N(u) = 1$ thus we deduce $N(u) = a^2 + 19b^2 = 1$ which gives necessarily $u = \pm 1$, then we have that $x + \sqrt{-19} = \left(a + b\sqrt{-19}\right)^3 = (a^3 - 3 \cdot 19ab^2) + (3a^2b - 19b^3)i$, by identification and enumerating the cases for $b = \pm 1$, we have that $3a^2 = 20$ which is impossible, and $a^2 = 6$ which is impossible too, thus there are no solutions. But notice that $7^3 = 18^2 + 19$ is a solution, which is contradictory, which comes from the fact that we assumed $\mathbb{Z}\left[\sqrt{-19}\right]$ has the properties $\mathbb{Z}[i]$, but $\mathbb{Z}\left[\sqrt{-19}\right]$ is not even a UFD since $35 = 5 \cdot 7 = \left(4 + \sqrt{-19}\right)\left(4 - \sqrt{-19}\right)$, we can take a better ring that contains $\mathbb{Z}\left[\sqrt{-19}\right]$, which is $\mathbb{Z}\left[\sqrt{-19}\right] \subset \mathbb{Z}\left[\frac{1+\sqrt{-19}}{2}\right]$ called the ring of integers of $\mathbb{Q}\left(\sqrt{-19}\right)$.

> **Exercise 1.1:** *Show that this implies that the only solutions for $y^3 = x^2 + 19$ are $y = 7$ and $x = \pm 18$.*

But it is not necessary in general that the ring of integers of a number field is a UFD. But, it has a unique factorization of prime ideals in rings of integers. Not all ideals are principal.

- **Problem 4:** Pell equation, let $d > 0$ consider the equation $x^2 - dy^2 = 1$, again, we take it in $\mathbb{Z}\left[\sqrt{d}\right]$, $\left(x + \sqrt{d}y\right)\left(x - \sqrt{d}y\right) = 1$, the norm $x + \sqrt{d} \mapsto x^2 - dy^2$, thus the problem of finding solutions to the Pell equation is equivalent to finding the units of the norm, almost all, if we take for example $d = 3$, we have that $x^2 - 3y^2 = -1$ is impossible since $\operatorname{mod} 4$ we obtain $x^2 + y^2 \equiv 3 \bmod 4$, therefore in this case, we have that the set of solutions is exactly the units. Since $(2, 1)$ is a solution then $2 + \sqrt{3} \in \mathbb{Z}\left[\sqrt{3}\right]^\star$ and thus it has infinitely many units so infinitely many solutions, we have that $\langle -1 \rangle \times \langle 2 + \sqrt{3} \rangle \subset \mathbb{Z}\left[\sqrt{3}\right]^\star$, more specifically, $\langle -1 \rangle \times \langle 2 + \sqrt{3} \rangle = \mathbb{Z}\left[\sqrt{3}\right]^\star$.

Theorem (Dirichlet Unit Theorem): *For any quadratic ring $\mathbb{Z}\left[\sqrt{d}\right]$, with $\mathbb{Z}\left[\sqrt{d}\right]^\star = \langle -1 \rangle \times \langle \varepsilon_d \rangle$ where ε_d is a fundamental unit with infinite order.*

Exercise 2: *Let $p \equiv 1 \bmod 4$ prime, show that $y = \left(\frac{p-1}{2}\right)!$ satisfies $y^2 \equiv -1 \bmod p$.*

Exercise 3:

1. *Show that $\mathbb{Z}\left[\sqrt{-2}\right]$ is a Euclidean domain, $N : a + b\sqrt{-2} \mapsto a^2 + 2b^2$.*
2. *Find the solutions for $y^3 = x^2 + 2$.*

Exercise 4: *Show that for $k \in \mathbb{Z}$ the sum of two integer squares if and only if $\forall p \mid k, p \equiv 3 \bmod 4$ then p appears with even exponent in the factorization of n.*

Consider the ring $R = \mathbb{Z}\left[\frac{1+\sqrt{-19}}{2}\right] = \mathbb{Z}[\alpha]$, it is not a Euclidean domain, that is, any function $h : R \setminus \{0\} \to \mathbb{Z}_{\geq 0}$ that satisfies $\forall a, b \in R, \exists! q, r \in R, a = bq + r, r = 0 \vee h(r) < h(b)$. Suppose by contradiction that h exists, we have the minimal polynomial of α is $f(X) = X^2 - X + 5$, to find $R^\star$ we consider the norm on R as $N : R \to \mathbb{Z}, a + b\alpha \mapsto a^2 + 2ab + 5b^2, x \in R^\star \Leftrightarrow N(x) = 1 \Leftrightarrow (a + b)^2 + 4b^2 = \pm 1 \Leftrightarrow a = \pm 1, b = 0$ thus $R^\star = \{\pm 1\}$. Now take $x \in R \setminus (R^\star \cup \{0\})$, with $h(x)$ minimal, look at $R/(xR)$, notice that by the Euclidean division, we get a surjection $R \twoheadrightarrow R/xR$. For any $\beta \in R, \exists q, r \in R, \beta = qx + r$, if $r = 0$ then $\overline{\beta} = 0 \in R/xR$, else if $r \neq 0$ then $h(r) < h(x)$ hence $r \in R^\star, \overline{\beta} = \overline{r} \in R/xR$. Therefore, $R/xR \cong \left\{\overline{0}, \overline{r}\right\} \Rightarrow \#(R/xR) \leq 3 \Rightarrow R/xR$ is either $\mathbb{F}_2$ and $\mathbb{F}_3$, we have that $\overline{\alpha} \in R/xR, \overline{\alpha}^2 - \overline{\alpha} + 5 = 0$ but $f(X)$ is irreducible in $\mathbb{F}_3[X]$.

Chapter 2

Number Fields

Definition 1 (Number Field): *A field K is said to be a number field if the extension K/Q is finite.*

Definition 2 (Number Ring): *Any subring of a number field is said to be a number ring.*

Theorem 3: *For any number field K, there exists a primitive element α, such that $K = \mathbb{Q}(\alpha) = \mathbb{Q}[X]/(f)$ with f the minimal polynomial of α.*

Example:

1. Quadratic extensions of $\mathbb{Q}$: the primitive element is canonical, $[K : \mathbb{Q}] = 2$ then there exists a unique square free integer d such that $K = \mathbb{Q}\left(\sqrt{d}\right)$.
2. Cyclotomic extensions: Let $\mathbb{Q} \subset \overline{\mathbb{Q}}$ an algebraic closure of $\mathbb{Q}$, fix $n \in \mathbb{N}_{>0}$, and we take the roots of unity $X^n - 1$ in $\overline{\mathbb{Q}}$, there are n of them, and μ_n is a cyclic group of order n, any generator is called a primitive n^{th} root of unity ζ_n, their minimal polynomial is called the n^{th} cyclotomic polynomial $\Phi_n \in \mathbb{Z}[X]$, that is

$$\Phi_n(X) = \prod_{\zeta_n \text{ primitive}} (X - \zeta_n) \in \overline{\mathbb{Q}}[X]$$

and we have that $\deg \Phi_n = \varphi(n) = \#(\mathbb{Z}/n\mathbb{Z})^\star$.

n	$\Phi_n(X)$
1	$X - 1$
2	$X + 1$
3	$X^2 + X + 1$
4	$X^2 + 1$
5	$X^4 + X^3 + X^2 + X + 1$
6	$X^2 - X + 1$
7	$X^6 + X^5 + \dots + X + 1$
8	$X^4 + 1$
9	$X^6 + X^3 + 1$
10	$X^4 - X^3 + X^2 - X + 1$

Notice that if n is prime then $\Phi_n(X) = X^{n-1} + \dots + X + 1$, also, all the coefficients of Φ_n are ± 1. But it is false, the first counterexample is Φ_{105} because $105 = 3 \cdot 5 \cdot 7$.

2.1. Galois Theory

Definition 4 (Galois Extension): *Let K be a field extension of $\mathbb{Q}$, we say that $K/\mathbb{Q}$ is a Galois extension if the minimal polynomial of any $\alpha \in K$ splits in $K[X]$, equivalently, K is splitting field of a polynomial $P \in \mathbb{Q}[X]$.*

Example:

1. Quadratic extensions are Galois.
2. Cyclotomic extensions are Galois given that it is the splitting field of $X^n - 1$.
3. $\mathbb{Q} \subset \mathbb{Q}\left(\sqrt[5]{2}\right)$, is not Galois since $\mathbb{Q}\left(\sqrt[5]{2}\right) \subset \mathbb{R}$ but there are complex roots of $X^5 - 2$.

Theorem 5 (Galois Theorem): *Let $K/\mathbb{Q}$ be a Galois extension, there is an inclusion reversing bijection:*

$$\begin{aligned} \{\text{subfield of } K\} &\longleftrightarrow \{\text{subgroups of } \mathrm{Gal}(K/\mathbb{Q})\} \\ L &\rightarrow \mathrm{Gal}(K/L) \\ K^H &\leftarrow H \end{aligned}$$

where

$$\mathrm{Gal}(K/\mathbb{Q}) = \{\sigma : K \to K \text{ automorphism} \mid \sigma|_{\mathbb{Q}} = \mathrm{id}|_{\mathbb{Q}}\}$$
$$K^H = \{x \in K \mid \forall \sigma \in H, \sigma(x) = x\}$$

and we have that $[L : \mathbb{Q}] = (\mathrm{Gal}(K/\mathbb{Q}) : \mathrm{Gal}(K/L))$, $|H| = [K : K^H]$ *and we have that* H *is normal in* $\mathrm{Gal}(K/\mathbb{Q})$ *if and only if* K^H *is Galois.*

Example:

1. Consider $K = \mathbb{Q}\left(\sqrt[3]{5}, \zeta_3\right)$, we have that $[K : \mathbb{Q}] = 6$, it is Galois since it is the splitting field of $X^3 - 5$ over $\mathbb{Q}$, we have then $\#\,\mathrm{Gal}(K/\mathbb{Q}) = 6$, we compute the elements of $\mathrm{Gal}(K/\mathbb{Q})$, we have
$$\tau : \begin{cases} \sqrt[3]{5} \mapsto \sqrt[3]{5} \\ \zeta_3 \mapsto \zeta_3^2 \end{cases}$$
$$\sigma : \begin{cases} \sqrt[3]{5} \mapsto \zeta_3 \sqrt[3]{5} \\ \zeta_3 \mapsto \zeta_3 \end{cases} \qquad \tau^2 : \begin{cases} \sqrt[3]{5} \mapsto \zeta_3 \sqrt[3]{5} \\ \zeta_3 \mapsto \zeta_3^2 \end{cases}$$
$$\sigma^2 : \begin{cases} \sqrt[3]{5} \mapsto \zeta_3^2 \sqrt[3]{5} \\ \zeta_3 \mapsto \zeta_3 \end{cases} \qquad \tau^3 : \begin{cases} \sqrt[3]{5} \mapsto \zeta_3^2 \sqrt[3]{5} \\ \zeta_3 \mapsto \zeta_3^2 \end{cases}$$
It is easy to calculate that $\sigma\tau \neq \tau\sigma$, thus $\mathrm{Gal}(K/\mathbb{Q}) = S_3$.
2. **Cyclotomic Fields:** Let ζ_n be a primitive element of unity. Let $f \in \mathrm{Gal}(\mathbb{Q}(\zeta_n)/\mathbb{Q})$, then $f : \zeta_n \mapsto \zeta_n^a$ with $\gcd(a, n) = 1$, and we have that
$$\mathrm{Gal}(\mathbb{Q}(\zeta_n)/\mathbb{Q}) \to (\mathbb{Z}/n\mathbb{Z})^\star$$
$$(\zeta_n \mapsto \zeta_n^a) \mapsto a \bmod n$$
an isomorphism, for example taking $n = 7$, we have that $\mathrm{Gal}(\mathbb{Q}(\zeta_7)/\mathbb{Q}) = (\mathbb{Z}/7\mathbb{Z})^\star = C_6$, thus there are two subgroups C_2 and C_3, which correspond to the subfields $\mathbb{Q}(\zeta_7 + \zeta_7^{-1})$ and $\mathbb{Q}(\zeta_7 + \zeta_7^2 + \zeta_7^4)$ respectively. We have that $[\mathbb{Q}(\zeta_7 + \zeta_7^2 + \zeta_7^4) : \mathbb{Q}] = 2$ thus $\mathbb{Q}(\zeta_7, \zeta_7^2 + \zeta_7^4) = \mathbb{Q}\left(\sqrt{d}\right)$, to find d, we consider the action of $\zeta_n \mapsto \zeta_n^{-1}$ on $\zeta_7 + \zeta_7^2 + \zeta_7^4$ and we get $\mathrm{Min}(\zeta_7 + \zeta_7^2 + \zeta_7^4) = (X - (\zeta_7 + \zeta_7^2 + \zeta_7^4))(X - (\zeta_7^3 + \zeta_7^5 + \zeta_7^6)) = X^2 - X + 2$ by having the heart to calculate it.

2.2. Ideals In Number Rings

Definition 6 (Operations On Ideals): *Let R be a ring and I, J be ideals.*
- *Addition:* $I + J = \{x + y \mid x \in I, y \in J\}$.
- *Multiplication:* $IJ = \{\sum x_i y_i \mid x_i \in I, y_i \in J\}$

Example:

- Let $R = \mathbb{Z}\left[\sqrt{-5}\right]$ and $I = \left(2, 1 + \sqrt{-5}\right)$, we have
$$\begin{aligned} I^2 &= \left(2, 1 + \sqrt{-5}\right)\left(2, 1 + \sqrt{-5}\right) \\ &= \left(4, 2\left(1 + \sqrt{-5}\right), -4 + 2\sqrt{-5}\right) \\ &= \left(4, 2 + 2\sqrt{-5}, 6\right) = (2) \end{aligned}$$
but I is not principal, suppose by contradiction that there exists $\alpha = a + b\sqrt{-5} \in R$ such that $I = (\alpha)$, we have that $(2) = I^2 = (\alpha)^2 = (\alpha^2)$ thus $(2) = (\alpha^2) \Rightarrow \alpha^2 = 2u$ with $u \in R^\star = \{\pm 1\}$, so $\alpha^2 = \pm 2 \Rightarrow \left(a + b\sqrt{-5}\right)^2 = a^2 - 5b^2 + 2ab\sqrt{-5} = \pm 2 \Rightarrow ab = 0 \Rightarrow a = 0$ or $b = 0$ which is a contradiction.

Definition 7 (Coprime Ideals): *Let R be a ring and I, J be two ideals, we say that I and J are coprime if $I + J = R$.*

Example:

- If $I = (x)$ and $J = (y)$ then $I + J = R \Leftrightarrow \exists a, b \in R, ax + by = 1$.
- If I, J are coprime then for any $n \geq 1, I^n, J^n$ are coprime. Let I, J coprime, then $R = I + J \Rightarrow R = R^{2n} = (I + J)^{2n}$ which is equal to
$$\underbrace{I^{2n}}_{\subset I^n} + \dots + \underbrace{I^{n+1}J^{n-1}}_{\subset I^n} + \underbrace{I^n J^n}_{\subset I^n + J^n} + \underbrace{I^{n-1}J^{n+1}}_{\subset J^n} + \dots + \underbrace{J^{2n}}_{\subset J^n} \subset I^n + J^n$$
thus $R = I^n + J^n$ so I^n, J^n are coprime.

Theorem 8: *Let I, J, Z be ideals of R, I and J are coprime, suppose that $IJ = Z^n$ then we have that $I = (I + Z)^n$ and $J = (J + Z)^n$.*

Proof sketch: $(I+Z)^n = I^n + I^{n-1}Z + \cdots + Z^n = I(I^{n-1} + I^{n-2}Z + \cdots + Z^{n-1} + J) = IR = I$.

Example: Consider $R = \mathbb{Z}[-19]$, we have $\left(18+\sqrt{-19}\right)\left(18-\sqrt{-19}\right) = 7^3$ then $\left(18+\sqrt{-19}\right)_R \left(18-\sqrt{-19}\right)_R = (7)_R^3$, we have then

$$\left(18+\sqrt{-19}\right) = \left[\left(18+\sqrt{-19}\right) + (7)\right]^3 = \left(18+\sqrt{-19}, 7\right)^3$$

which is not principal.

2.2.1. Fractional Ideals

Let R be a number ring, and $K = \mathrm{Frac}(R)$ the field of fractions or R.

Definition 9 (Fractional Ideal): *Let $I \subset K, I \neq \{0\}$, we say that I is fractional if it an additive subgroup of K that satisfies:*
- *$\forall r \in R, rI \subset I$. (Ideal of R)*
- *$\exists x \in K^\star, xI \subset R$.*

Example:
1. Ideals of R are fractional ideals (integral ideals).
2. $\forall x \in K^\star, xR = (x)_R$ is a fractional ideal (principal fractional ideals).
3. Take $R = \mathbb{Z}\left[\sqrt{-5}\right], I = R + R\frac{1+\sqrt{-5}}{2}$ is a fractional non-principal ideal.

Definition 10 (Ideal Quotient): *Let I, J be two fractional ideals of R, we define the quotient of I over J as*

$$I : J = \{x \in K \mid xJ \subset I\}$$

Proposition 11: *Let I, J be two fractional ideals of R, then $I : J$ is also a fractional ideal of R.*

Proof. $(I : J) \neq \{0\}$ since if we take $a \in I \setminus \{0\}$ and $b \in K^\star : bJ \subset R$ then $abJ \subset aR \subset I \Rightarrow ab \in (I : J)$. It is clear that it is an additive subgroup of K. Take $x \in (I : J)$ we have that $xJ \subset I \Rightarrow rxJ \subset rI \subset I \Rightarrow r(I : J) \subset (I : J)$. Let $c \in K^\star$ such that $cI \subset R$, and $d \in J \setminus \{0\}$, take $x \in (I : J), cdx \in R, dx \in I \Rightarrow cdx \in R \Rightarrow cd(I : J) \subset R$. Thus, $(I : J)$ is a fractional ideal. ■

Definition 12 (Inverse/Multiplier Ring): *Let I be a fractional ideal, then*
- *Inverse ideal: $I^{-1} = R : I = \{x \in K \mid xI \subset R\}$*
- *Multiplier ring: $r(I) = I : I = \{x \in K \mid xI \subset I\}$*

Proposition 13: *Let R be a ring and I a fractional ideal of R.*
- *$II^{-1} \subset R$.*
- *If $I = (a)$ then $I^{-1} = (a^{-1})$ and $II^{-1} = R$.*
- *The multiplier ring $r(I)$ is a subring of K.*

Proof.
- $\forall x \in I^{-1}, xI \subset R \Rightarrow I^{-1}I = II^{-1} \subset R$.
- Suppose $I = (a) = aR$, let $x \in I^{-1}$ then $xaR = xI \subset R \Leftrightarrow xaR \subset R \Leftrightarrow xR \subset a^{-1}R \Leftrightarrow x \in a^{-1}R$ and hence $I^{-1} = a^{-1}R = (a^{-1})$, also, $II^{-1} = (aR)(a^{-1}R) = aa^{-1}R = R$.
- Do it. ■

Example: Notice that in general, we do not necessarily have that $II^{-1} = R$. Consider $R = \mathbb{Z}\left[\sqrt{5}\right]$ and $I = \left(2, 1+\sqrt{5}\right)$, we claim that $II^{-1} \neq R$. Suppose that $II^{-1} = R$, we have

$$I^2 = \left(4, 6+2\sqrt{5}, 2+2\sqrt{5}\right) = \left(4, 2+2\sqrt{5}\right) = 2\left(2, 1+\sqrt{5}\right) = 2I$$

A, multiplying both sides by I^{-1} we get $I = IR = (2)$ then I is principal, but we have that I is indeed not principal, contradiction, so $II^{-1} \neq R$.

Definition 14 (Invertible Ideal): *A fractional ideal I is said to be invertible if $II^{-1} = R$. The set of invertible ideals is denoted $I(R)$.*

Proposition 15: *$I(R)$ forms a group under ideal multiplication.*

2.2.2. Factoring Into Prime Ideals

Let $n \in \mathbb{Z}$, from the fundamental theorem of arithmetic, we have that n can be decomposed into $n = \prod_i p_i^{e_i}$ unique up to units and order. We can deduce that $(n) = \prod_i (p_i)^{e_i}$. The question now is by considering any ideal $I \subset R$, can we decompose in a similar way the ideal I into a product $I = \prod \mathfrak{p}_i^{e_i}$ where $\mathfrak{p}_i$ are prime ideals.

Exercise 16: *Let R be a number ring, and let $I \neq \{0\}$ be an ideal of R, prove that I contains a positive integer.*

Let $\alpha \in I \setminus \{0\}$, given that $\alpha \in R$ then α is algebraic over $\mathbb{Q}$, then there exists $(a_i)_{i=1}^{n} \in \mathbb{Z}$, $a_n\alpha^n + \cdots + a_1\alpha + a_0 = 0$, moving elements around we get

$$a_n\alpha^n + \cdots + a_1\alpha = -a_0$$

we have that the LHS is in the ideal since $(\alpha) \subset I$, and thus $-a_0 \in I \Rightarrow |a_0| \in I$, and $|a_0|$ is a positive integer.

Let R be a number ring and $I \neq \{0\}$ an ideal of R, and take a prime ideal $\mathfrak{p}_i$ such that $I \subset \mathfrak{p}_i$, from the exercise we have that there exists $n \in I$ thus $n \in \mathfrak{p}_i$ but $\mathfrak{p}_i$ is prime then it contains a prime number p_i therefore $\mathfrak{p}_i \cap \mathbb{Z} = p_i\mathbb{Z}$.

Theorem 17: *Every ideal I in a number ring R has finite index.*

Proof. Let R be a number ring and $K = \operatorname{Frac}(R)$ a number field, $[K : \mathbb{Q}] = t$. From the exercise, we have that there exists a positive integer $a \in I$. We have that $(a) \subset I$, then there exists a surjective map $R/aR \twoheadrightarrow R/I$, thus it is enough to prove that R/aR is finite.

Lemma: *Let G be a group, such that any finitely generated subgroup of G is of order at most n, then G is a finite of order at most n.*

Let M be a finitely generated subgroup of R as an additive group. M has no elements of finite order given that $\operatorname{char}(R) = \operatorname{char}(K) = 0$, thus M is a free abelian group, and since $[K : \mathbb{Q}] = t$ then $\operatorname{rank}(M) \leq t$. Any finitely generated subgroup of R/aR is the image of a finitely generated subgroup of R by the surjection $R \twoheadrightarrow R/aR$, and thus any finitely generated subgroup of R/aR has order $a^k \leq a^t$, thus we conclude that R/aR has finite order. ∎

Corollary 18: *Every prime ideal of a number ring is maximal.*

Proof. Let $\mathfrak{p}$ be a prime ideal of R, we have that $R/\mathfrak{p}$ is finite and an integral domain, thus $R/\mathfrak{p}$ is a field, and thus $\mathfrak{p}$ is a maximal ideal in R. ∎

In general, prime ideal factorization does not hold even if I is invertible, the problem is the lack of invertibility of prime ideals, which offers two solutions:

1. Avoid non-invertible prime ideals.
2. Find rings R where every prime ideal is invertible (Dedkind rings).

Exercise 19:

1. *Show that the ideals of $\left(18 + \sqrt{-19}\right)$ and $\left(18 - \sqrt{-19}\right)$ are coprime in $\mathbb{Z}\left[\sqrt{-19}\right]$.*
2. *Show that $\left(18 + \sqrt{-19}, 7\right)$ is a maximal ideal of index 7 in $\mathbb{Z}\left[\sqrt{-19}\right]$.*
3. *Show that if $I, J \subset R$ are coprime then $IJ = I \cap J$.*
4. *Compute $r(I)$ for $I = \left(2, \sqrt{5} + 1\right)$ in $\mathbb{Z}\left[\sqrt{5}\right]$.*
5. *Show that $I = \left(2, 1 + \sqrt{-5}\right)$ is invertible in $\mathbb{Z}\left[\sqrt{-5}\right]$.*
6. *Show that any ideal $I \neq \{0\}$ of a number ring contains a positive integer.*
7. *Show that a number ring has only finitely many ideals I satisfying for some fixed bound B, $\#(R : I) \leq B$.*

Let K be a number field, we have that $K = \mathbb{Q}(\alpha)$ for some primitive element α. We have that $\mathbb{Z} \subset \mathbb{Q}$ thus $\mathbb{Z} \subset K$ and $\alpha \in K$ so $\mathbb{Z}[\alpha] \subset K$.

$$\begin{tikzcd}
\mathbb{Z} \arrow[r, "\subset"] \arrow[d, hook] & \mathbb{Q} \arrow[d, hook, "n"] \\
\mathbb{Z}[\alpha] \arrow[r, "\subset"] & K
\end{tikzcd}$$

$\mathbb{Z}[\alpha]^\star$ can be infinite, but it is finitely generated, as stated in the Dirichlet unit theorem.

Consider now the field $K = \mathbb{Q}[X]/(f(X))$ where $f(X) = X^2 + 5$, and the ring $R = \mathbb{Z}\left[\sqrt{-5}\right]$, take the ideal $I = (6) = 6\mathbb{Z}\left[\sqrt{-5}\right]$ which we want to factorize, we start by finding all prime ideals $\mathfrak{p}$ such that $(6) \subset \mathfrak{p} \subset \mathbb{Z}\left[\sqrt{-5}\right]$. We have that $\mathbb{Z}\left[\sqrt{-5}\right] = \mathbb{Z} + \mathbb{Z}\sqrt{-5}$,

$$R/I = \frac{\mathbb{Z} + \mathbb{Z}\sqrt{-5}}{6\mathbb{Z} + 6\mathbb{Z}\sqrt{-5}} \cong (\mathbb{Z}/6\mathbb{Z}) \times (\mathbb{Z}/6\mathbb{Z})$$

and therefore $\# R/I = 36 = 2^2 \cdot 3^2$, since any ideal of a number field is maximal, then R/I is a field, we have 4 possible fields it can be, either a field of order $2, 3, 2^2, 3^2$ so R/I can have either characteristic 2, 3. Suppose that the characteristic is 3, and consider a morphism from $\mathbb{Z}\left[\sqrt{-5}\right]$ to $\mathbb{F}_3$ defined as $\varphi_3 : a + b\sqrt{-5} \mapsto a + b\alpha \bmod 3$, for φ_3 to be a ring morphism we need α to be a root of $f \bmod 3$ which is $X^2 - 1 = (X - 1)(X + 1) \bmod 3$ so we take $\alpha = \pm 1$, the kernel of those morphisms would give ideals of $\mathbb{Z}\left[\sqrt{-5}\right]$ which are $\mathfrak{p}_3 = \ker\left(a + b\sqrt{-5} \mapsto a + b \bmod 3\right) = \left(3, 1 - \sqrt{-5}\right)$ and $\mathfrak{q}_3 = \ker(a + b\sqrt{-5} \mapsto a - b \bmod 3 = \left(3, 1 + \sqrt{-5}\right)$, notice now that $\mathfrak{p}_3\mathfrak{q}_3 = \left(3, 1 - \sqrt{-5}\right)\left(3, 1 + \sqrt{-5}\right) = \left(9, 3 + 3\sqrt{-5}, 3 - 3\sqrt{-5}, 6\right) = (3)$, now suppose that the field is of characteristic 2, then $\varphi_2 : a + b\sqrt{-5} \mapsto a + b \bmod 2$ since $X^2 + 5 = (X + 1)^2 \bmod 2$, $\ker \varphi_2 = \left(2, 1 + \sqrt{-5}\right) = \mathfrak{p}_2$, we have that $\mathfrak{p}_2^2 = \left(4, 2 + 2\sqrt{-5}, -4 + 2\sqrt{-5}\right) = (2)$. We have that $6 = 2 \cdot 3 \Rightarrow (6) = (2)(3) = \mathfrak{p}_2^2\mathfrak{p}_3\mathfrak{q}_3$, notice that we can get other decomposition of 6 inside of $\mathbb{Z}\left[\sqrt{-5}\right]$ by considering other orders of the product of those ideals, for example taking $(6) = \mathfrak{p}_2\mathfrak{p}_3\mathfrak{p}_2\mathfrak{q}_3 = \left(6, 1 + \sqrt{-5}\right)\left(6, 1 - \sqrt{-5}\right)$, so 6 can be decomposed also into $\left(1 + \sqrt{-5}\right)\left(1 - \sqrt{-5}\right)$.

---------------------- Stevenhagen Course Unfinished ----------------------

Chapter 3

Rings Of Integers

Definition 1 (Integral Element/Ring Of Integers): *Let K be a number field, and $\beta \in K$. We say that β is an integral element in K if there exists a monic polynomial $f \in \mathbb{Z}[T]$ such that $f(\beta) = 0$, denote the set of all integral elements of K as O_K which is called the ring of integers.*

Proposition 2: *Let $x \in K$, the following statements are equivalent:*

1. *x is integral.*
2. *the minimal polynomial of x, $f_{x,\mathbb{Q}}$ is in $\mathbb{Z}[X]$.*
3. *there exists a finitely generated subgroup $M \subset K$ such that $xM \subset M$.*

Proof.

- 1. $\Rightarrow$ 2. By Gauss lemma.
- 2. $\Rightarrow$ 3. Let $f_{x,\mathbb{Q}} = T^n + a_{n-1}T^{n-1} + \cdots + a_0 \in \mathbb{Z}[T]$, consider $M = \mathbb{Z}x + \mathbb{Z}x^2 + \cdots + \mathbb{Z}x^{n-1}$, we have that $x^n = xx^{n-1} = -a_{n-1}x^{n-1} - \cdots - a_0 \in M$ then $xM \subset M$.
- 3. $\Rightarrow$ 1. Consider $M = \mathbb{Z}e_1 + \cdots + \mathbb{Z}e_n$ with $e_i \in K$, by hypothesis, we have that $\forall i, xe_i \in M$ thus there exists $a_{i,j} \in \mathbb{Z}, xe_i = a_{i,1}e_1 + \cdots + a_{i,n}e_n$, we can write it in the following way

$$\begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{n1} & \cdots & a_{nn} \end{pmatrix} \begin{pmatrix} e_1 \\ \vdots \\ e_n \end{pmatrix} = x \begin{pmatrix} e_1 \\ \vdots \\ e_n \end{pmatrix} \Rightarrow (A - x1_n) \begin{pmatrix} e_1 \\ \vdots \\ e_n \end{pmatrix} = 0$$

  thus x is a root of characteristic polynomial of A, which is monic and has integer coefficients, thus x is integral. ■

Proposition 3: *O_K is a ring.*

Proof.

1. Let $x, y \in O_K$, we want to prove that $xy, x + y \in O_K$, let $M = \mathbb{Z}e_1 + \cdots + \mathbb{Z}e_m, xM \subset M$ and $N = \mathbb{Z}f_1 + \cdots + \mathbb{Z}f_n, yN \subset N$. Take $L = \sum_{i,j} \mathbb{Z}e_i f_j$, we have that $(xy)(e_i f_j) = (xe_i)(yf_j) \in L$ and thus $xyL \subset L$ and L is finitely generated thus $xy \in O_K$. ■

Combined with Kummer-Dedekind, we can produce elements in the ring of integers: let $\alpha \in O_K \Rightarrow \mathbb{Z}[\alpha] \subset O_K$, suppose that $\mathfrak{p} = (p, g(\alpha))$ is singular ($f = gq + ps$, $q(\alpha), s(\alpha) \in \mathfrak{p}$), we have that $\frac{1}{p}q(\alpha)$ is in the multiplier ring of $\mathfrak{p}$:

$$\frac{1}{p}q(\alpha)(pR + g(\alpha)R) = q(\alpha)R + s(\alpha)R \in \mathfrak{p}$$

but notice that $\mathfrak{p} = pR + g(\alpha)R = \alpha(\mathbb{Z} + \mathbb{Z}\alpha + \cdots + \mathbb{Z}\alpha^{-1} + g(\alpha)(\mathbb{Z} + \mathbb{Z}\alpha + \cdots + \mathbb{Z}\alpha^{n-1}) \subset \mathfrak{p}$

Example:

1. Let $K = \mathbb{Q}(\alpha)$, $\alpha = \sqrt[3]{10}$, $f_\alpha = X^3 - 10 \Rightarrow \alpha \in O_K \Rightarrow \mathbb{Z}[\alpha] \subset O_K$. $\Delta f_\alpha = -3^3 \cdot 2^2 \cdot 5^2$
   - $p = 2$: $f_\alpha \bmod 2 = X^3 \to \mathfrak{p}_2 = (2, \alpha)$ and $2^2 \nmid 10 \Rightarrow \mathfrak{p}_2$ is regular.
   - $p = 5$: $f_\alpha \bmod 5 = X^3 \to \mathfrak{p}_5 = (5, \alpha)$ which is regular since $5^2 \nmid 10$.
   - $p = 3$: $f_\alpha \bmod 3 = (X - 1)^3 \to \mathfrak{p}_3 = (3, \alpha - 1)$, $f_\alpha(X) = X^3 - 10 = X^3 - 1 - 9 = (X - 1)(X^2 + X + 1) - 9 \Rightarrow 3^2 \mid 9$ and thus $\mathfrak{p}_3$ is singular, from Kummer-Dedekind, we have that $\beta = \frac{\alpha^2+\alpha+1}{3} \in O_K$, we can verify that by calculating the minimal polynomial $f_\beta = X^3 - X^2 - 3X - 3$.

   We can deduce from the formula $\Delta f = [O_K : \mathbb{Z}[\alpha]]^2 \Delta_K$, thus $[O_K : \mathbb{Z}[\alpha]] = 3$ and $\Delta_K = -3 \cdot 2^2 \cdot 5^2$, we deduce that $\mathbb{Z}[\alpha] \subsetneq \mathbb{Z}[\alpha, \beta] = O_K$, we claim that $O_K = \mathbb{Z}[\beta]$, by Kummer-Dedekind, we have that $(\alpha - 1)\beta = 3 \Rightarrow \alpha = \frac{3}{\beta} + 1$, from the minimal polynomial of β we have that $\beta(\beta^2 - \beta - 3) = \beta^3 - \beta^2 - 3\beta = 3 \Rightarrow \frac{\beta}{3} = \beta^3 - \beta^2 - 3\beta$ hence $\alpha = \beta^2 - \beta - 2 \in \mathbb{Z}[\beta]$.

   We expect that all rings of integers are of the form $\mathbb{Z}[\alpha]$, which is not the case, and we call those who happen to be of that form to be monogenic.
2. Let $K = \mathbb{Q}(\alpha)$ with $f_\alpha = X^3 - X + 8$, we have that $\Delta f_\alpha = -2^2 \cdot 431$, we have that $\mathbb{Z}[\alpha \subset O_K$, only possible singular prime is 2, $f_\alpha \equiv$

$X(X-1)^2 \bmod 2 \to \mathfrak{p}_2 = (2, \alpha)$ and $\mathfrak{p}_2' = (2, \alpha - 1)$, $\mathfrak{p}_2$ is regular, $f_\alpha = X^3 - X + 8 = (X-1)(X^2+X) + 8$, $2^2 \mid 8$ then $\mathfrak{p}_2'$ is singular. From Kummer-Dedekind we have that $\beta = \frac{\alpha^2+\alpha}{2} \in O_K \setminus \mathbb{Z}[\alpha]$ with $f_\beta = X^3 - X^2 + 6X - 8$, from the formula we have that $[O_K : \mathbb{Z}[\alpha]] = 2$ then $\mathbb{Z}[\alpha, \beta] = O_K$, the question now is $\mathbb{Z}[\alpha, \beta] = \mathbb{Z}[\beta]$ like before?

- $p = 2$: $f_\beta \equiv X^2(X-1) \bmod 2$, then in $\mathbb{Z}[\beta]$, 2 primes $\mathfrak{q}_2 = (2, \beta)$ and $\mathfrak{q}_2' = (2, \beta - 1)$, $\mathfrak{q}_2'$ is regular, and $\mathfrak{q}_2$ is singular, thus $\mathbb{Z}[\beta] \neq O_K$

Suppose now that there is $\gamma \in O_K$ such that $O_K = \mathbb{Z}[\gamma]$, we have that $(\alpha - 1)\beta = -4 \Leftrightarrow (\alpha - 1)\beta = 0 \in \mathbb{F}_2$, which gives 3 ring homomorphisms, $(\alpha, \beta) \mapsto (1, 0), (\alpha, \beta) \mapsto (1, 1), (\alpha, \beta) \mapsto (0, 0)$, in fact all possibilities occur:

$$2O_K = (2, \alpha - 1, \beta)(2, \alpha - 1, \beta - 1)(2, \alpha, \beta)$$

if by contradiction O_K is monogenic, $O_K = \mathbb{Z}[\gamma]$, it has at most 2 homomorphisms to $\mathbb{F}_2$ which is a contradiction.

3.1. Trace And Norm

Definition 4 (Trace/Norm): *Let K a number field and $x \in K$, define the map $m_x : K \mapsto K, \alpha \mapsto x\alpha$ which is a $\mathbb{Q}$-linear map.*
- *Characteristic polynomial of x is $\operatorname{char}(m_x) \in \mathbb{Q}[X]$.*
- *Norm of x is $\det(m_x)$, denoted $N_{K/Q}(x) \in \mathbb{Q}$.*
- *Trace of x is $\operatorname{tr}(m_x)$, denoted $\operatorname{Tr}_{K/Q}(x) \in \mathbb{Q}$.*

Proposition 5: *Let K be a number field, let $x \in K$ with a characteristic polynomial $\operatorname{char}(m_x) = T^n + a_{m-1}T^{m-1} + \dots + a_0$. We have that*
- $N_{K/Q}(x) = (-1)^m a_0 \in \mathbb{Q}$.
- $\operatorname{Tr}_{K/Q}(x) = -a_{m-1} \in \mathbb{Q}$.
- $N_{K/Q}(xy) = N_{K/Q}(x) N_{K/Q}(y)$.
- $\operatorname{Tr}(x + y) = \operatorname{Tr}(x) + \operatorname{Tr}(y)$.

Example:
- Let $K = \mathbb{Q}(\sqrt{d})$, we have $N_{K/Q}(a + b\sqrt{d}) = a^2 - db^2$. Lets verify that this agrees with the new definition, $X = a + b\sqrt{d}$, consider the basis $(1, \sqrt{d})$ then $x \cdot 1 = a + b\sqrt{d}$ and $x\sqrt{d} = db + \sqrt{d}$ then $m_x = \begin{pmatrix} a & db \\ b & a \end{pmatrix}$, which has norm $a^2 - db^2$, and trace $2a$.
- Let $K = \mathbb{Q}(\sqrt[4]{2})$, and $x = \sqrt{2}$, we have a basis $(1, \alpha, \alpha^2, \alpha^3)$ with $\alpha = \sqrt[4]{2}$, notice that $x = \alpha^2$, we have

$$\begin{cases} \alpha^2 \cdot 1 & = \alpha^2 \\ \alpha^2 \cdot \alpha & = \alpha^3 \\ \alpha^2 \cdot \alpha^2 & = 2 \\ \alpha^2 \cdot \alpha^3 & = 2\alpha \end{cases} \Rightarrow m_x = \begin{pmatrix} 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}$$

we have that $\operatorname{char}(x), \operatorname{char}(m_x) = T^4 - 4T^2 + 4 = (T^2 - 2)^2$, let now $L = \mathbb{Q}(\sqrt{2})$, $x \in L$, we have that $\operatorname{char}(x) = T^2 - 2$.

3.2. Embeddings

Let K be a number field, by the primitive root theorem, there exists a primitive element α such that $K = \mathbb{Q}(\alpha) = \mathbb{Q}[X]/(f)$ with $f = f_\alpha^{\mathbb{Q}}$. K can be embedded in $\mathbb{C}$ $n = [K : \mathbb{Q}]$ different ways, the embeddings would be of the form $\Phi_i : \alpha - X \bmod f \mapsto \alpha_i$ where $\alpha_1, \dots, \alpha_n$ are the complex roots of α. The Φ_i for which $\Phi_i(K) \subset \mathbb{R}$ are called real embeddings, and the number of those embeddings is denoted r, and the other embeddings such that $\Phi_i(K) \not\subset \mathbb{R}$ we call them complex embeddings, which come in pairs $\Phi_i, \overline{\Phi_i}$, and we take s the number of pairs of complex embeddings. (r, s) is the signature of K.

We can consider all the embeddings at the same time, instead of looking to $\mathbb{Q}[X]/(f)$, we look for $\mathbb{C}[X]/(f)$, notice that $\mathbb{Q}[X]/(f)$ is a field while $\mathbb{C}[X]/(f)$ is a ring, but notice that in $\mathbb{C}[X]$ we have that $f = \prod(X - \alpha_i)$, thus by the CRT we have that $\mathbb{C}[X]/(f) \cong \prod_{i=1}^n \mathbb{C}[X]/(X - \alpha_i) \cong \prod_{i=1}^n \mathbb{C}$, thus the injective map from $\mathbb{Q}[X]/(f) \to \prod_{i=1}^n \mathbb{C}$ gives $\alpha = X \bmod f \mapsto (\alpha_1, \dots, \alpha_n) = (\varphi_1(\alpha), \dots, \varphi_n(\alpha))$, denote this map Φ

$$\Phi : \mathbb{Q}[X]/(f) \to \prod_{i=1}^n \mathbb{C}, \qquad \alpha \mapsto (\Phi_1(\alpha), \dots, \Phi_n(\alpha))$$

Theorem 6: *Let K be a number field and $x \in K$.*
- $N_{K/\mathbb{Q}}(x) = \prod_{i=1}^n \Phi_i(x)$.

- $\mathrm{Tr}_{K/\mathbb{Q}}(x) = \sum_{i=1}^{n} \Phi_i(x)$.
- $\text{char }(x) = \left(f_{K,\mathbb{Q}}\right)^{[K:\mathbb{Q}(x)]}$.